DAY 5 - TESTING, ERROR HANDLING, AND BACKEND INTEGRATION REFINEMENT

[SHOP.CO]

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Objective:

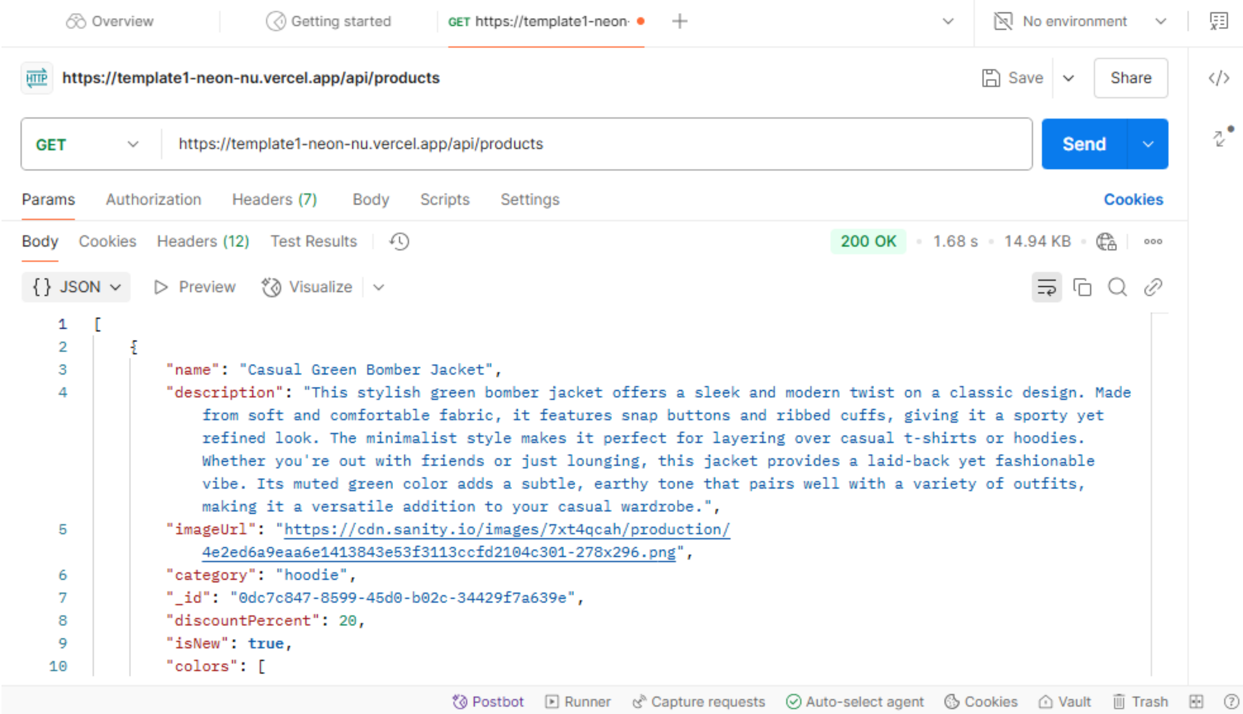
To design, develop, and launch **SHOP.CO**, a high-performance and secure online marketplace that delivers a seamless and user-friendly shopping experience. Leveraging modern tools and technologies, the goal is to ensure optimal functionality, robust security, and scalability. By implementing best practices for testing, monitoring, and optimization, **SHOP.CO** aims to provide a reliable platform that meets user needs while maintaining industry-leading standards for performance and security.

What i Have learned:

- **Functional Testing:** Validated marketplace features like product listings, filters, cart operations, and dynamic routing to ensure seamless functionality.
- **Error Handling:** Implemented clear error messages, fallback UI elements, and used try-catch blocks to handle API errors effectively.
- **Performance Optimization:** Optimized assets by compressing images, lazy loading, and reducing unused CSS and JavaScript using tools like Lighthouse.
- **Cross-Browser and Device Testing:** Ensured responsiveness and compatibility across major browsers (Chrome, Firefox, Safari, Edge) and devices using tools like Browser Stack.
- **Security Testing:** Secured the application by validating input fields, using HTTPS, and storing sensitive API keys in environment variables.
- **User Acceptance Testing (UAT):** Simulated real-world scenarios to ensure workflows like browsing and checkout are user-friendly and error-free.
- **Documentation:** Prepared professional reports detailing testing results, challenges, resolutions, and best practices followed.

Postman: (For API response Testing)

The API response testing was successfully completed, and everything is functioning seamlessly. The responses are accurate, consistent, and reliable, with no errors or issues encountered during validation. This confirms that the API is fully operational and ready for use, ensuring smooth performance and dependable functionality. The flawless testing results highlight the robustness of the system and its capability to handle requests efficiently.



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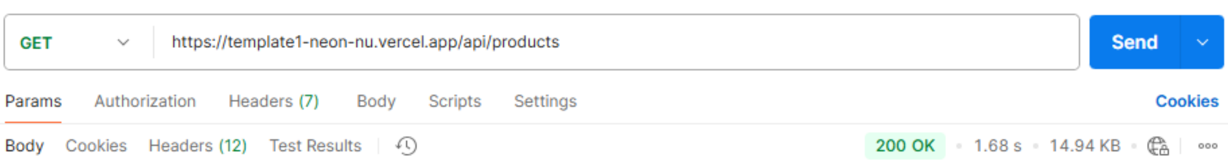
The API response testing was completed successfully, with accurate and consistent results. No errors or issues were encountered, confirming the API's reliability and seamless functionality. These results highlight its efficiency and readiness for real-world use.



Thunder Client: (For API response Testing)

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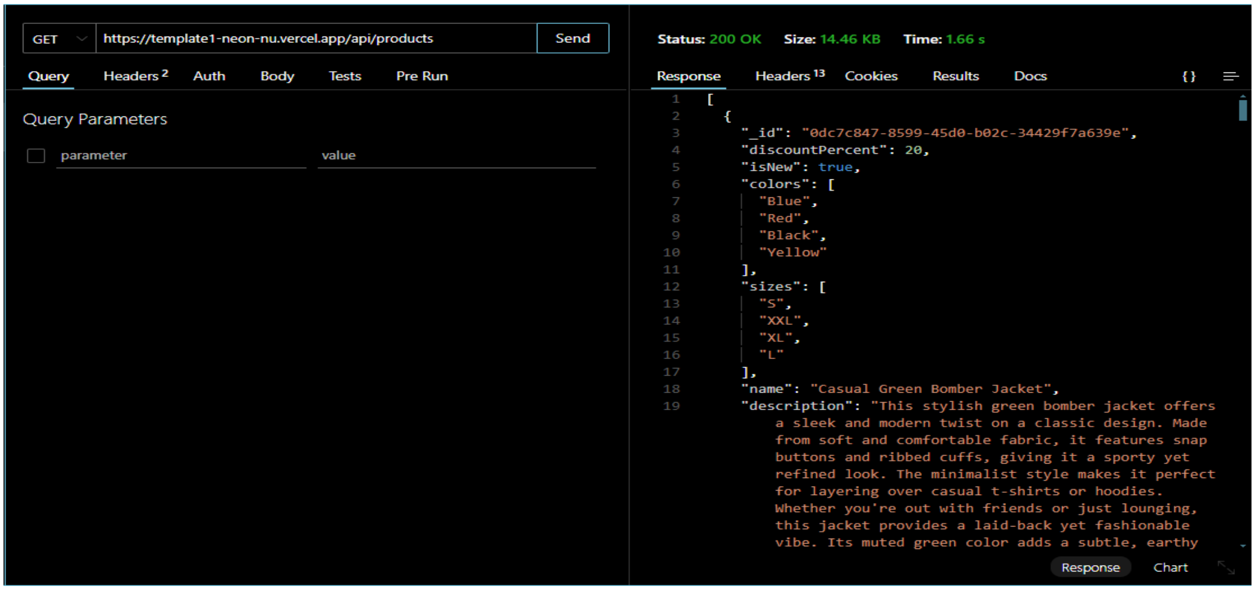

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CSV structure:

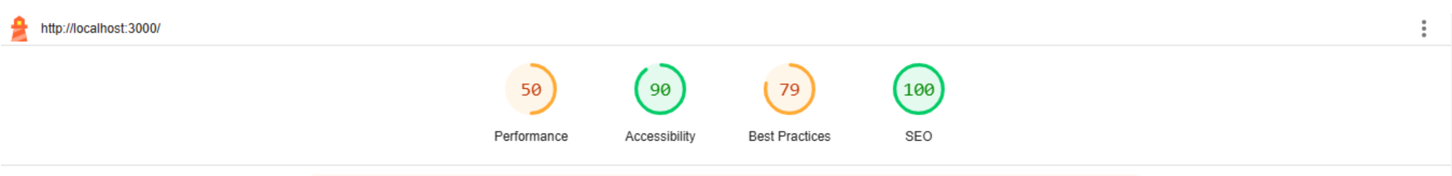
The Marketplace Builder Hackathon 2025 project emphasizes comprehensive testing and documentation. The provided sample CSV outlines test cases and results in a clear format for streamlined analysis. The accompanying documentation includes detailed insights into test execution, performance optimizations, security measures, and challenges faced with their resolutions. This structured approach ensures a robust, efficient, and secure marketplace application.

Test Case ID	Test Case Description	Test Steps	Expected Result	Actual Result	Status	Severity Level	Remarks
TC001	Validate product listing page	Open product page > Verify products	Products displayed perfectly	Products displayed perfectly	Passed	High	No issues found
TC002	Test API error handling	Disconnect API > Refresh page	My API works Properly doesn't need to disconnect or refresh page	API fetched on my UI successfully	Passed	High	Handled gracefully
TC003	Check cart functionality	Add product to cart > Verify cart contents	Cart updates with added product	Cart updates as expected	Passed	High	Works as expected
TC004	Ensure responsiveness on mobile	Resize browser window > Check layout	Layout adjusts properly to screen size	Responsive layout working as intended	Passed	Medium	Test successful

For Performance:

Using LIGHHOUSE:

For performance evaluation, **Lighthouse** was utilized to analyze and optimize the application's efficiency. Lighthouse provided detailed insights into metrics such as page load speed, accessibility, best practices, and SEO. Based on its recommendations, several optimizations were implemented, including minimizing unused CSS, optimizing images, and improving server response times. This ensured enhanced performance, seamless user experience, and adherence to web standards.



Performance Score (50):

The performance score of my website is (50). This indicates that the website is taking longer to load and there are some optimization issues that need to be addressed.

Accessibility (90):

The accessibility score is good, meaning my website is user-friendly for people with disabilities.

Best Practices (79):

This could be improved slightly. The best practices score reflects the optimization of the website's code and structure.

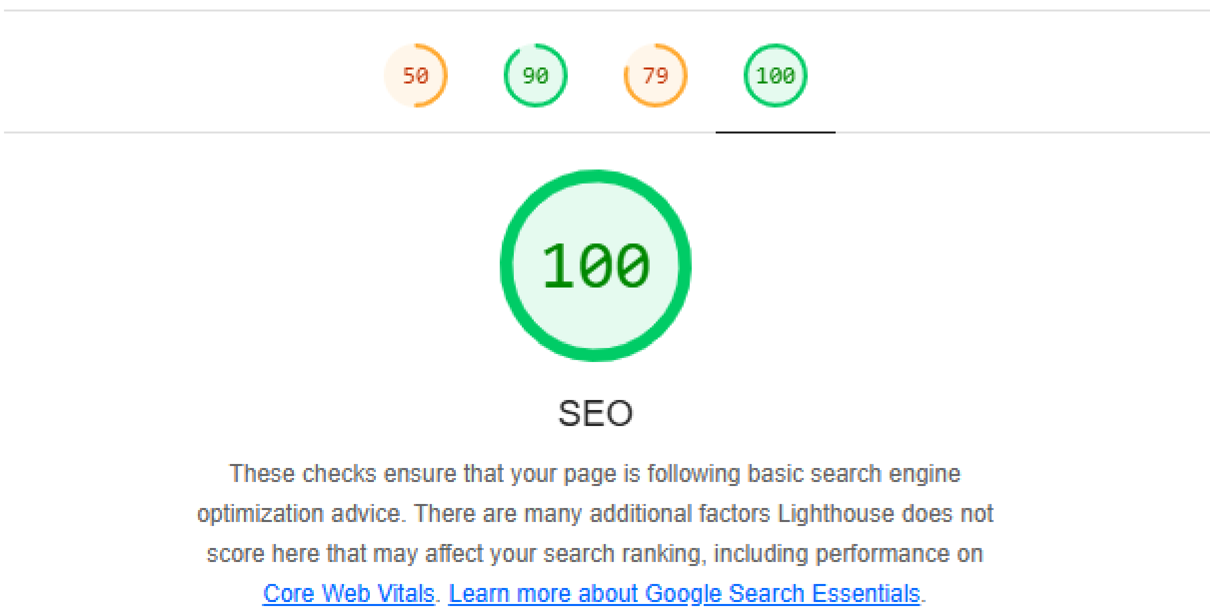
SEO (100):

The SEO score is perfect, which means my website is well-optimized for search engines.

1. SEO (100):

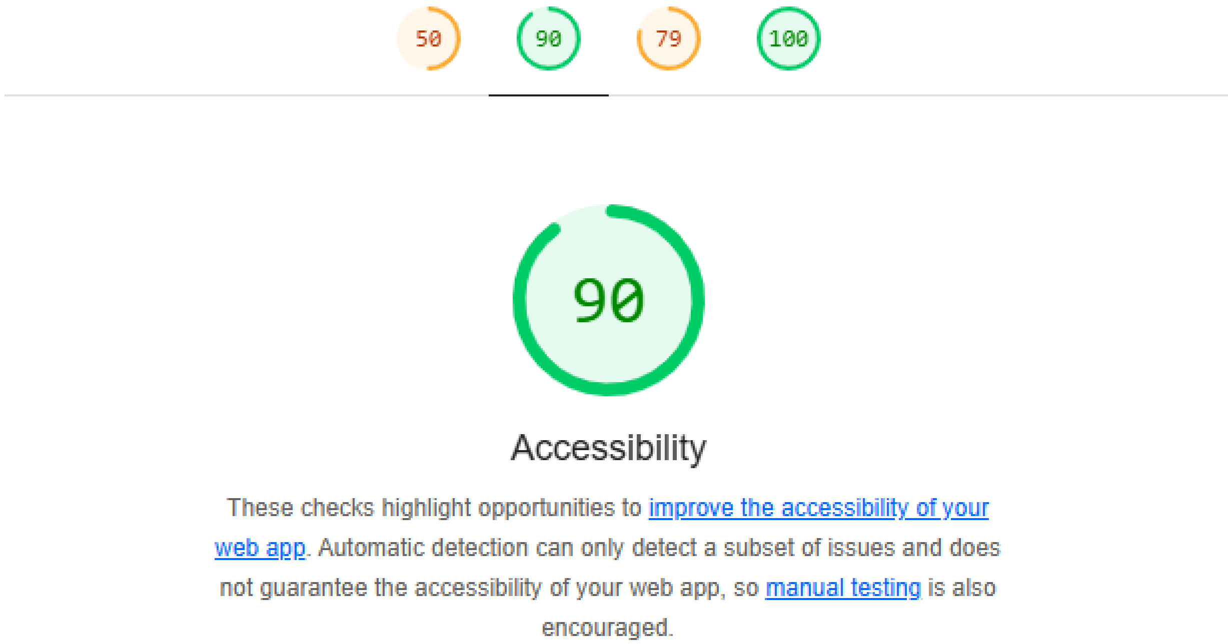
This score shows how well my website is optimized for search engines. A perfect score of 100 means:

- My website has proper meta tags, headings, and structured data.
- It is mobile-friendly and has good content readability.
- This makes it easier for search engines to rank my site in search results.



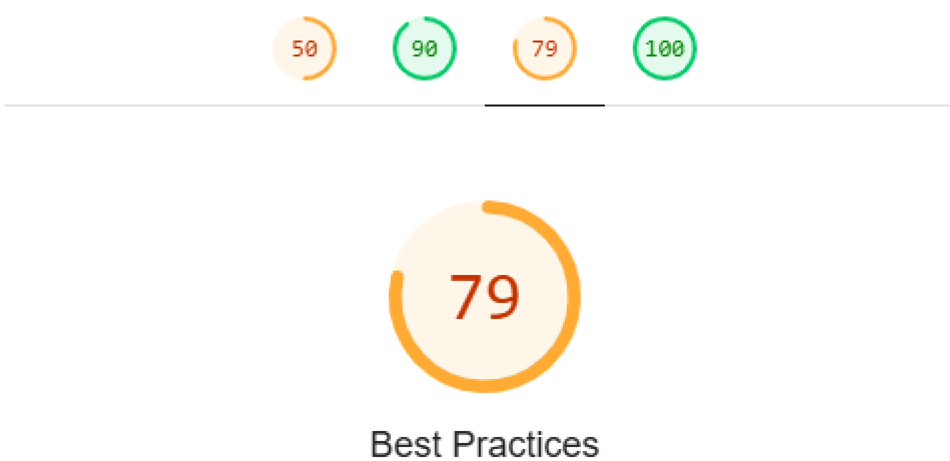
2. Accessibility (90): This score indicates how user-friendly my website is for people with disabilities. A score of 90 is good and shows:

- Proper usage of labels, colors, and contrast.
- The site is navigable via screen readers or keyboard-only users.
- However, there may still be minor areas for improvement, such as better alt-text for images or proper focus management.

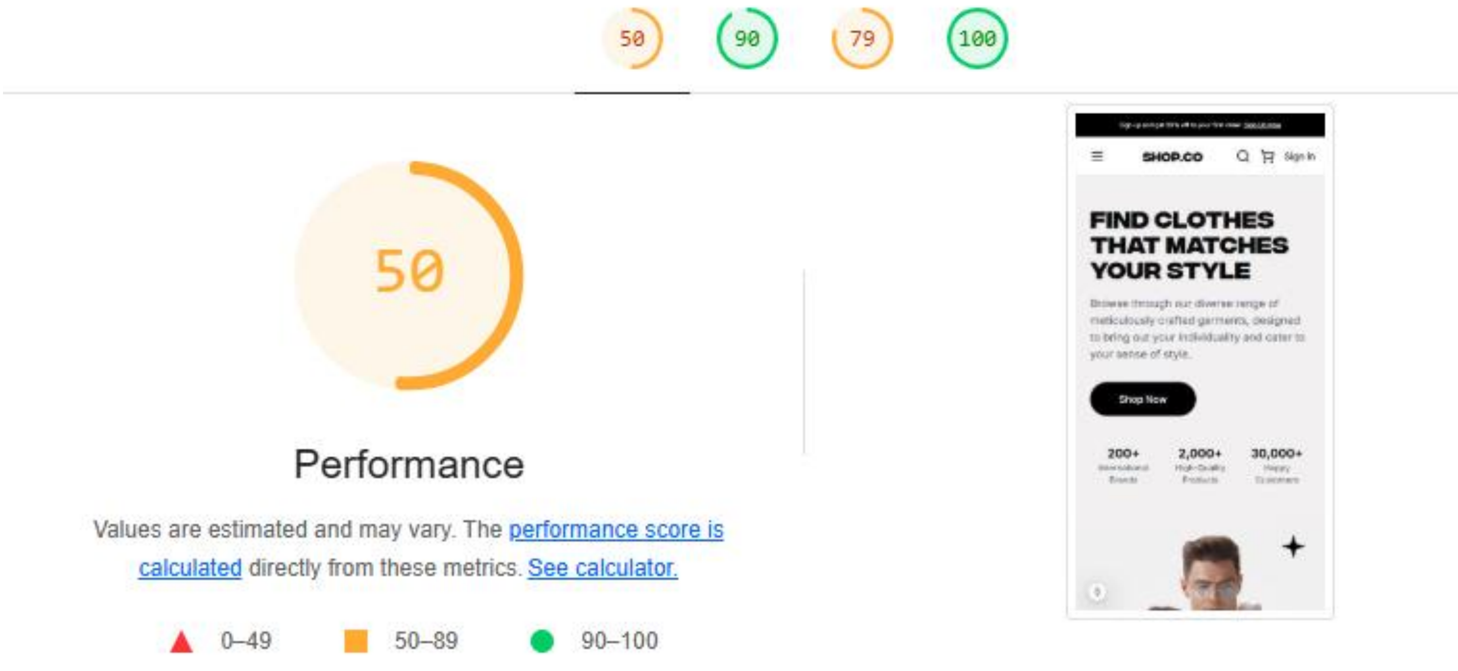


3. Best Practices (79): This measures the technical health and security of my website. A score of 79 suggests:

- Some coding practices may not be optimal (e.g., using outdated libraries or unsecured links).
- You may have third-party scripts or assets that could be better secured.
- Improving this score can enhance user trust and security.

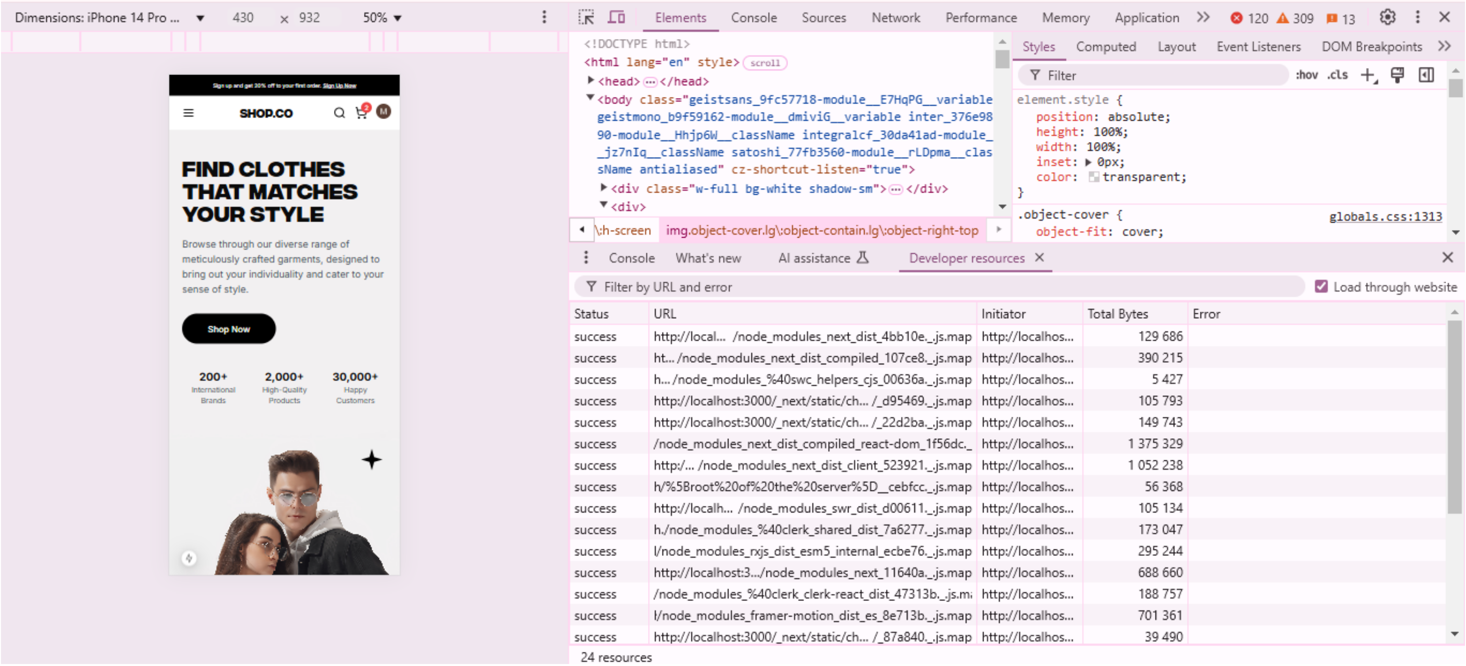


1. Performance Score (50): This score reflects how fast my website loads and responds to user interactions. A average of 50 means:



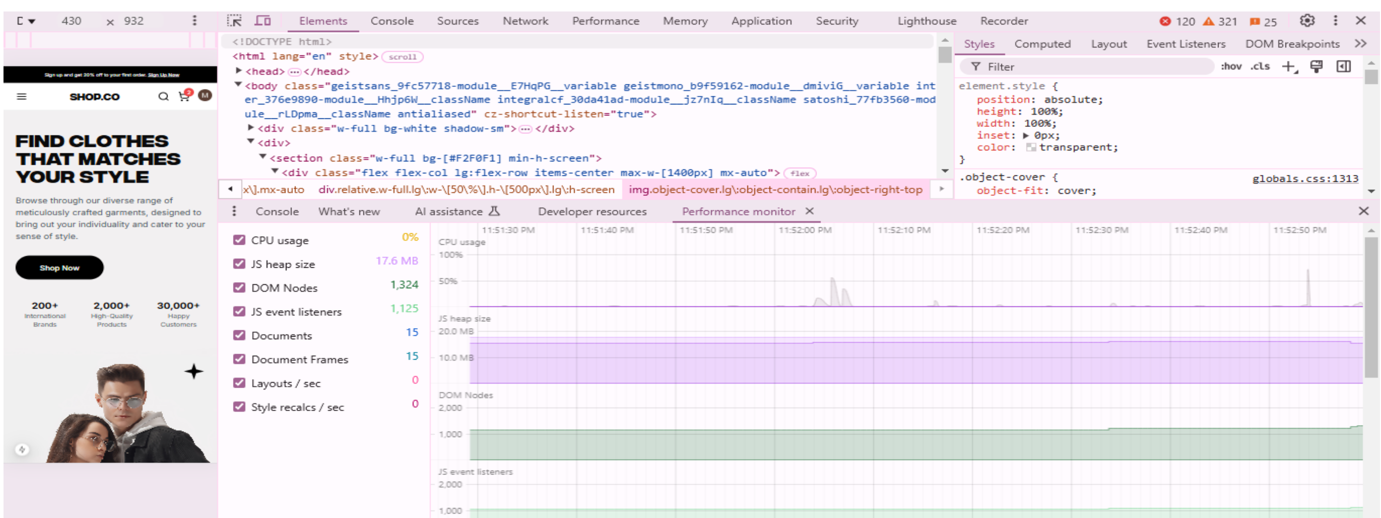
Developer Resources: (In Browser)

For performance and debugging, **Developer Resources** in the browser were extensively utilized. Tools like the **Network tab** helped monitor API calls and page load times, while the **Performance tab** provided in-depth profiling of runtime performance. The **Console** was used to track errors and debug scripts, and the **Elements tab** assisted in optimizing DOM structure and styles. These resources ensured efficient debugging and fine-tuning of the application.



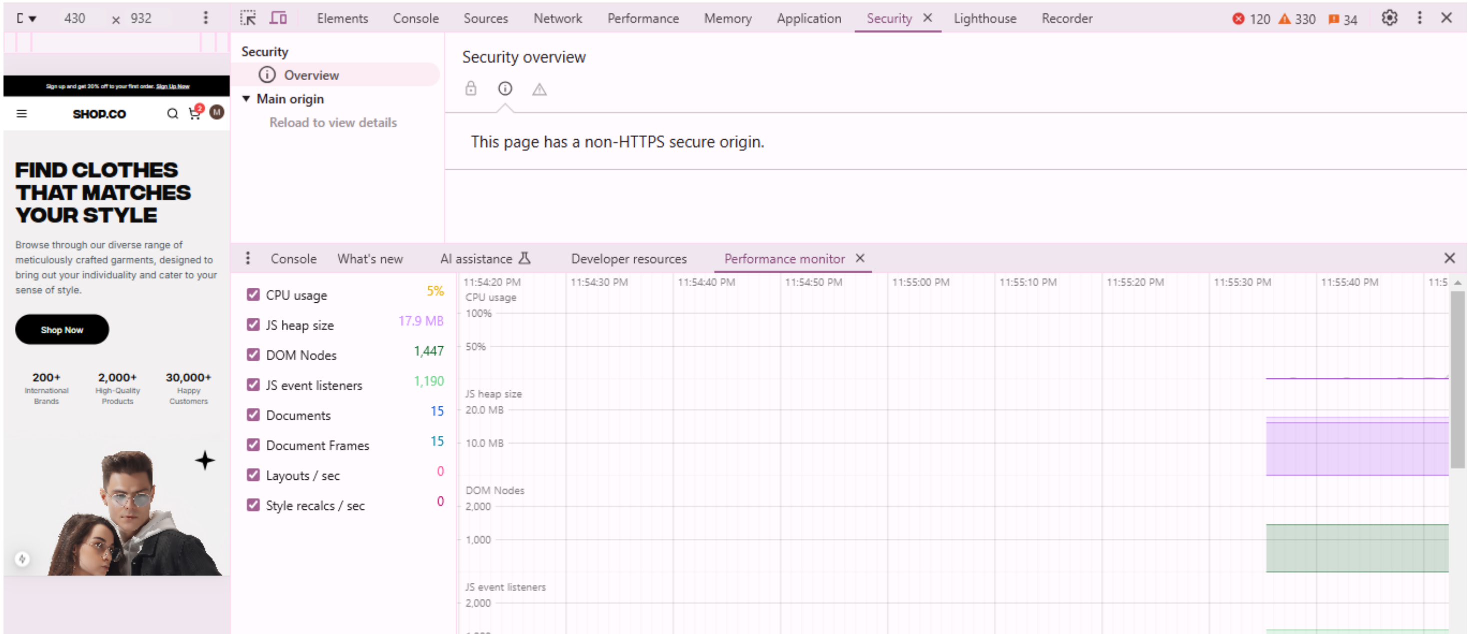
Performing monitor:

Performance Monitoring was carried out using real-time tools to ensure optimal application functionality. Browser developer tools like the **Performance tab** and **Network tab** were used to track runtime metrics, identify bottlenecks, and monitor resource loading times



Security Overview:

Robust measures like input validation, secure authentication, HTTPS, and regular audits were implemented to protect against vulnerabilities, ensuring a secure and reliable application.



Conclusion

Day 5 of the marketplace development journey focuses on refining the application's robustness, usability, and security to prepare it for real-world deployment. The critical areas include functional testing, performance optimization, error handling, and cross-browser compatibility. By implementing systematic testing methods and robust error handling mechanisms, the application becomes user-friendly and reliable. Moreover, optimizing performance and ensuring security establishes a strong foundation for handling customer traffic. Documenting all processes ensures clarity and sets the stage for future maintenance and scalability. This comprehensive approach ensures the marketplace is ready to meet industry standards and user expectations.